Project Title: Online Desktop Exam Simulator with Anti-Cheat System

Introduction: With the rapid growth of online education, ensuring exam integrity has become a critical challenge. Most existing solutions rely on browser-based systems, which are vulnerable to cheating techniques such as tab switching, screen recording, and external applications.

This project proposes a desktop-based online examination software that enforces a controlled exam environment while maintaining online connectivity for centralized data management. The system focuses on security, reliability, and real-world applicability.

Problem Statement:

Current online exam systems face the following issues:

- Dependence on browsers with limited control
- High risk of cheating via system-level access
- Poor monitoring of student behavior
- Internet instability affecting exam integrity

Objectives:

- To design a desktop-based online exam software
- To implement OS-level anti-cheat mechanisms
- To ensure secure client–server communication
- To provide automated evaluation and reporting
- To create a system suitable for real academic use

Scope of the Project:

Admin:

- Create and manage exams
- Control exam schedules and rules
- Monitor student activity and violations
- Access reports and results

Student:

- Secure login via desktop application
- Attempt exams in full-screen locked mode
- Automatic submission on timeout or violation
- Real-time activity monitoring

System Features:

- Role-based authentication
- Full-screen enforced exam mode
- Anti-cheat detection:
  - Alt+Tab / focus loss
  - Banned applications
  - Multiple login detection
- Auto evaluation of MCQs
- Detailed reports with integrity logs

Technology Stack:

- Desktop Application: Python (Tkinter / PyQt)
- Backend Server: Python (FastAPI / Flask)
- Database: MySQL
- Security: JWT, encrypted communication (conceptual)
- OS Monitoring: psutil, threading

Expected Outcome:

- Practical software engineering
- Security-aware design
- Real-world problem solving
- Academic and professional readiness

Future Enhancements:

- Facial recognition (optional)
- AI-based behavior analysis